DC TRANSIENT ANALYSIS

SUB - TOPICS

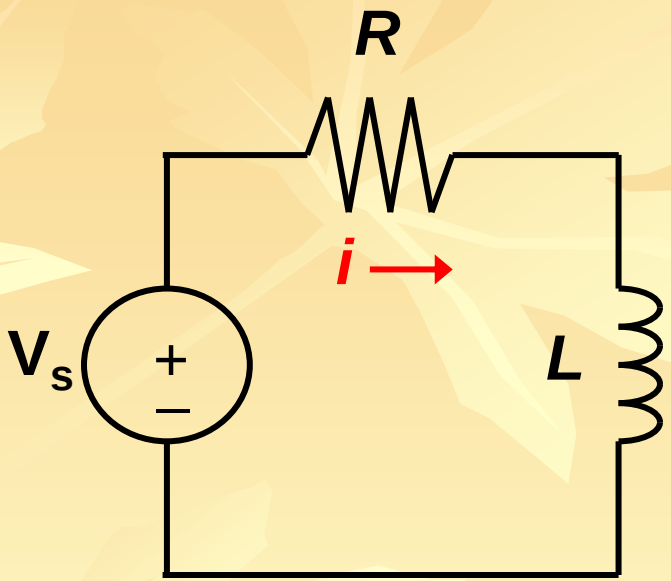
- NATURAL RESPONSE OF RL CIRCUIT
- NATURAL RESPONSE OF RC CIRCUIT
- STEP RESPONSE OF RL CIRCUIT
- STEP RESPONSE OF RC CIRCUIT

OBJECTIVES

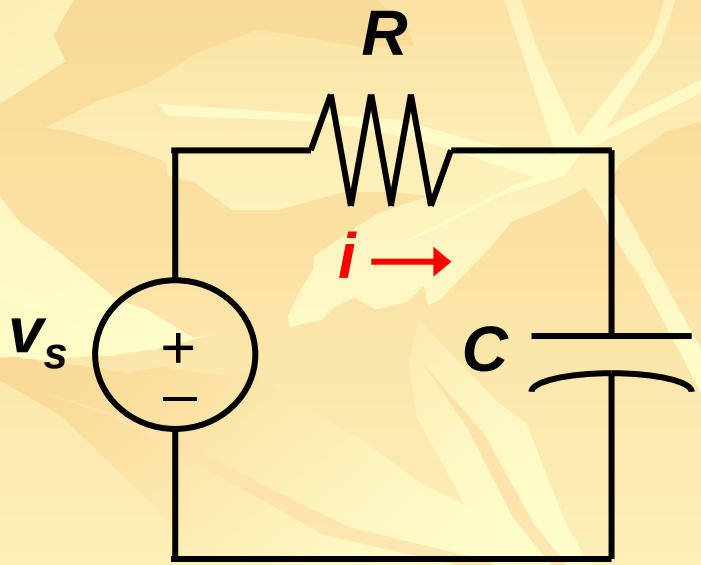
- To investigate the behavior of currents and voltages when energy is either released or acquired by inductors and capacitors when there is an abrupt change in dc current or voltage source.
- To do an analysis of natural response and step response of RL and RC circuit.

FIRST – ORDER CIRCUITS

- A circuit that contains only sources, resistor and inductor is called and RL circuit.
- A circuit that contains only sources, resistor and capacitor is called an RC circuit.
- RL and RC circuits are called first – order circuits because their voltages and currents are describe by first order differential equations.



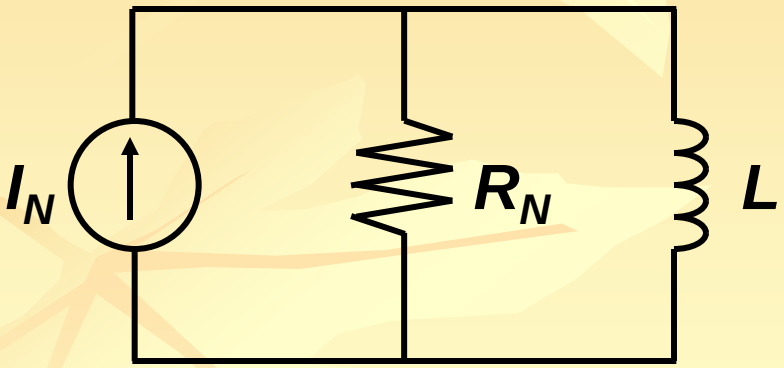
An RL circuit

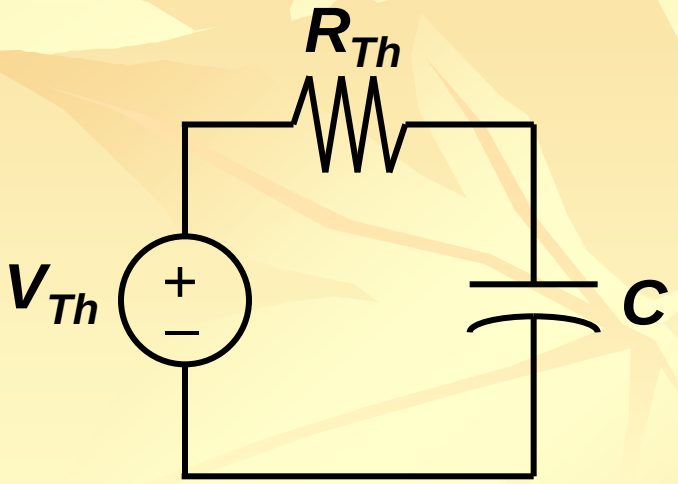


An RC circuit

Review (conceptual)

- Any first – order circuit can be reduced to a Thévenin (or Norton) equivalent connected to either a single equivalent inductor or capacitor.



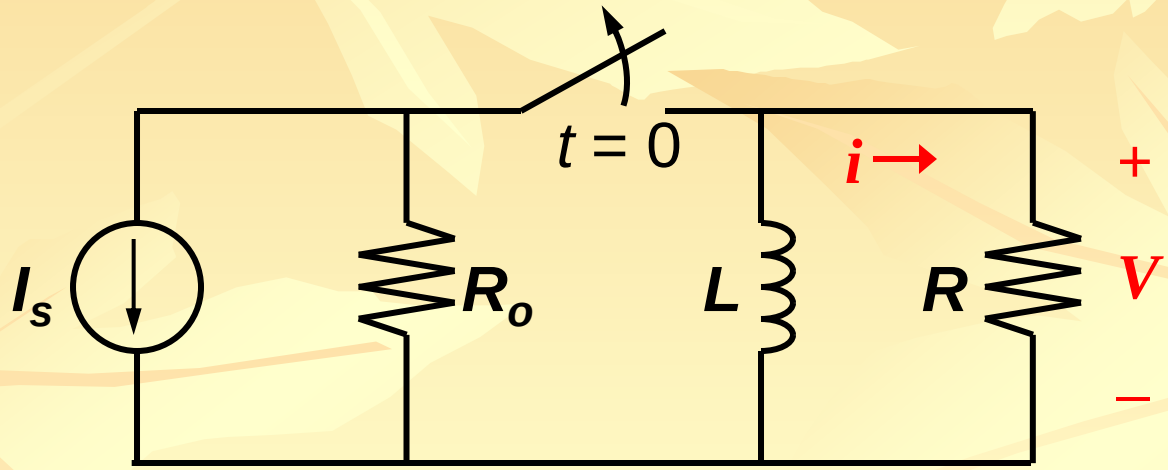


- In steady state, an inductor behave like a short circuit.
- In steady state, a capacitor behaves like an open circuit.

- The natural response of an RL and RC circuit is its behavior (i.e., current and voltage) when stored energy in the inductor or capacitor is released to the resistive part of the network (containing no independent sources)
- The steps response of an RL and RC circuits is its behavior when a voltage or current source step is applied to the circuit, or immediately after a switch state is changed.

NATURAL RESPONSE OF AN RL CIRCUIT

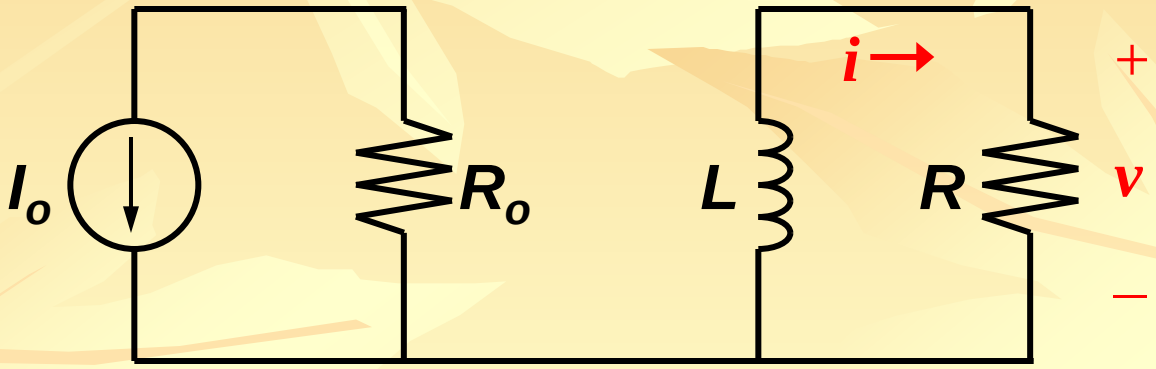
- Consider the following circuit, for which the switch is closed for $t<0$, and then opened at $t = 0$:



- The dc voltage V, has been supplying the RL circuit with constant current for a long time

Solving for the circuit

- For $t \leq 0$, $i(t) = I_o$
- For $t \geq 0$, the circuit reduce to



Notation:

- 0^- is used to denote the time just prior to switching.
- 0^+ is used to denote the time immediately after switching.

Continue…

- Applying KVL to the circuit:

$$v(t) + Ri(t) = 0 \qquad \text{——————} \quad \textbf{(1)}$$

$$L\frac{di(t)}{dt} + Ri(t) = 0 \qquad \text{——————} \quad \textbf{(2)}$$

$$L\frac{di(t)}{dt} = -Ri(t) \qquad \text{——————} \quad \textbf{(3)}$$

$$\frac{di(t)}{i(t)} = -\frac{R}{L}dt \qquad \text{——————} \quad \textbf{(4)}$$

Continue

- From equation (4), let say;

$$\frac{du}{u} = -\frac{R}{L}dv \quad \text{——————} \quad \textbf{(5)}$$

- Integrate both sides of equation (5);

$$\int_{i(t_o)}^{i(t)} \frac{du}{u} = -\frac{R}{L}\int_{t_o}^{t} dv \quad \text{——————} \quad \textbf{(6)}$$

- Where:
  - $i(t_o)$ is the current corresponding to time t_o
  - $i(t)$ ia the current corresponding to time t

Continue

- Therefore,

$$\ln\frac{i(t)}{i(0)} = -\frac{R}{L}t \qquad \text{(7)}$$

- hence, the current is

$$i(t) = i(0)e^{-(R/L)t} = I_0 e^{-(R/L)t}$$

Continue

- From the Ohm's law, the voltage across the resistor R is:

$$v(t) = i(t)R = I_0 \,\mathrm{Re}^{-(R/L)t}$$

- And the power dissipated in the resistor is:

$$p = v_R i(t) = I_0^2 \,\mathrm{Re}^{-2(R/L)t}$$

Continue

- Energy absorb by the resistor is:

$$w = \frac{1}{2} L I_0^2 (1 - e^{-2(R/L)t})$$

Time Constant, τ

- Time constant, τ determines the rate at which the current or voltage approaches zero.

- Time constant, $\tau = \frac{L}{R}$ (sec)

- The expressions for current, voltage, power and energy using time constant concept:

$$i(t) = I_0 e^{-t/\tau}$$

$$v(t) = I_0 \,\mathrm{Re}^{-t/\tau}$$

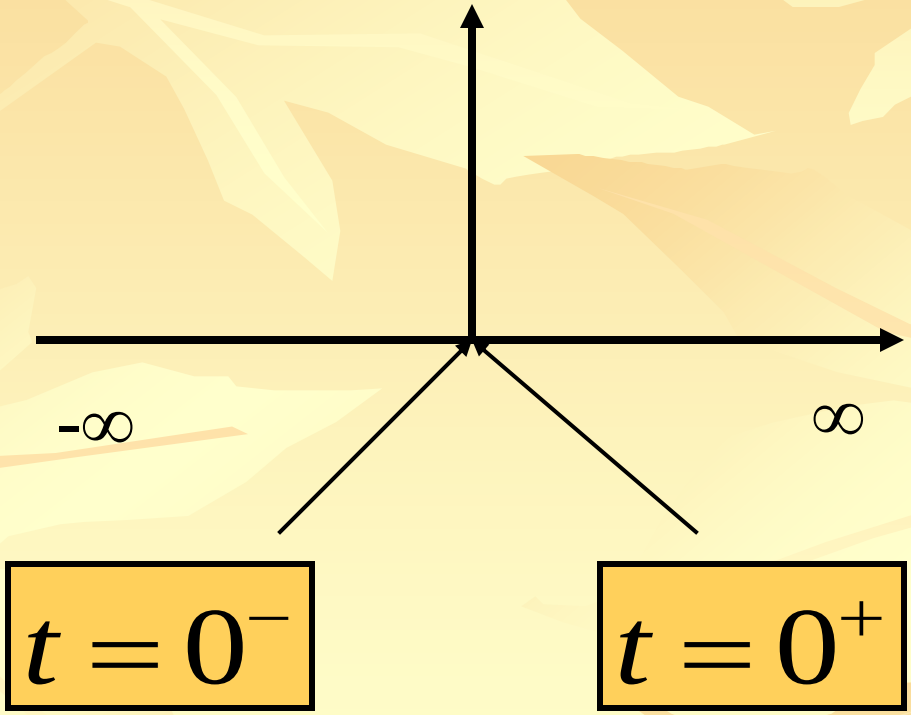
$$p = I_0^2 \,\mathrm{Re}^{-2t/\tau}$$

$$w = \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau})$$

Switching time

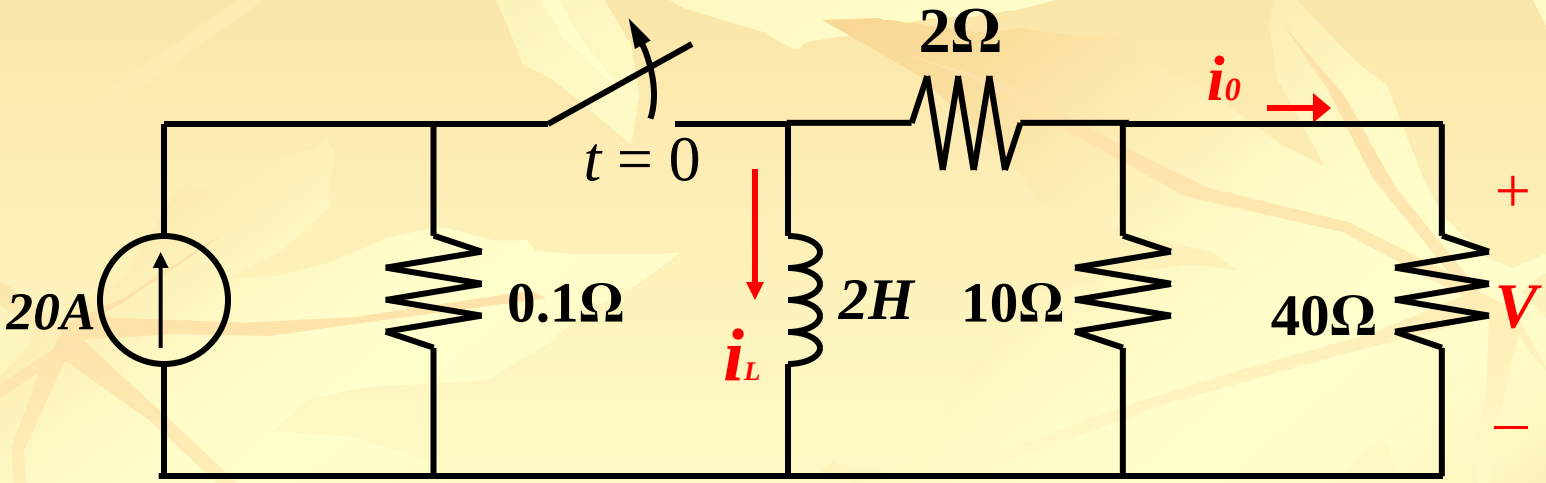
- For all transient cases, the following instants of switching times are considered.
- ✓ $t = 0^-$, this is the time of switching between $-\infty$ to 0 or time before.
- ✓ $t = 0^+$, this is the time of switching at the instant just after time t = 0s (taken as initial value)
- ✓ $t = \infty$, this is the time of switching between $t = 0^+$ to ∞ (taken as final value for step response)

- The illustration of the different instance of switching times is:



Example

- For the circuit below, find the expression of $i_o(t)$ and $V_o(t)$. The switch was closed for a long time, and at $t = 0$, the switch was opened.



Solution :

Step 1:

Find τ for $t > 0$. Draw the equivalent circuit. The switch is opened.

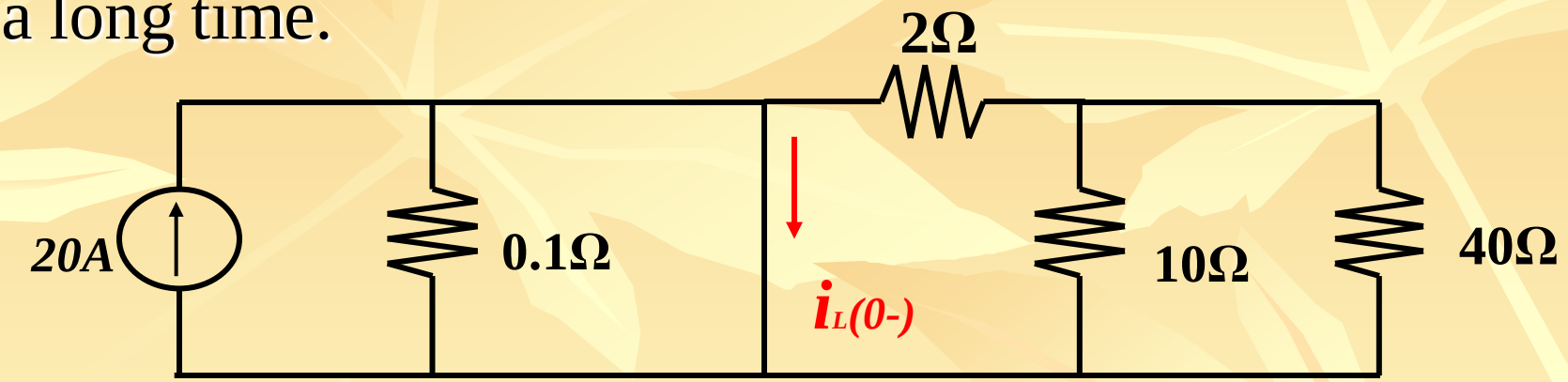
$$R_T = (2 + 10 // 40) = 10\Omega$$

So;

$$\tau = \frac{L}{R_T} = \frac{2}{10} = 0.2 \text{ sec}$$

Step 2:

At $t = 0^-$, time from $-\infty$ to 0^-, the switch was closed for a long time.

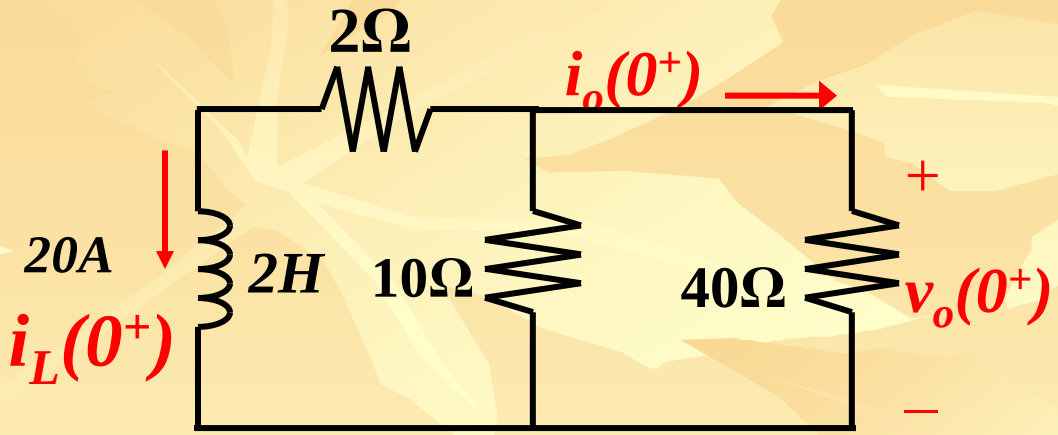


The inductor behave like a short circuit as it being supplied for a long time by a dc current source. Current 20A thus flows through the short circuit until the switch is opened.

Therefore; $i_L(0^-) = 20A$

Step 3:

At the instant when the switch is opened, the time $t = 0^+$,



The current through the inductor remains the same (continuous).

Thus, $i_L(0^+) = i_L(0^-) = 20A$ which is the initial current.

Only at this particular instant the value of the current through the inductor is the same.

Since, there is no other supply in the circuit after the switch is opened, this is the natural response case.

By using current division, the current in the 40Ω resistor is:

$$i_o = -i_L \frac{10}{10+40} = -4A$$

So,

$$\boxed{i_o(t) = -4e^{-5t}A}$$

Using Ohm's Law, the V_o is:

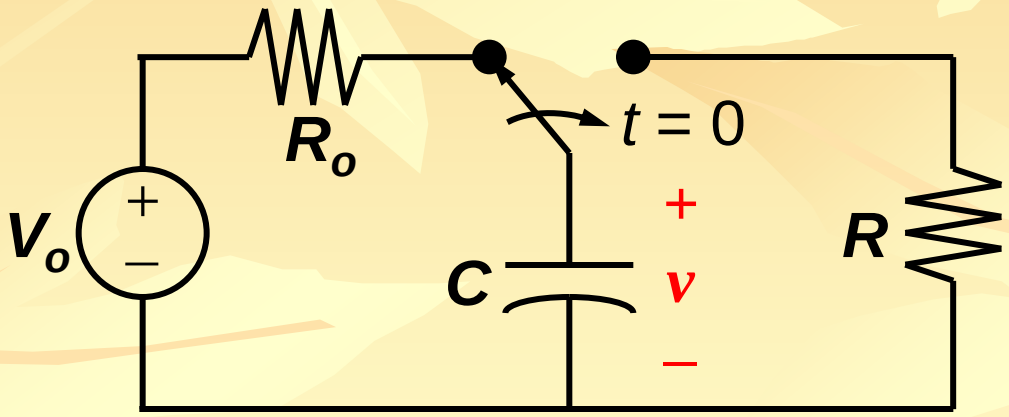
$$V_o(t) = -4 \times 40 = -160$$

So,

$$\boxed{V_0(t) = -160e^{-5t}}$$

NATURAL RESPONSE OF AN RC CIRCUIT

- Consider the following circuit, for which the switch is closed for $t < 0$, and then opened at $t = 0$:

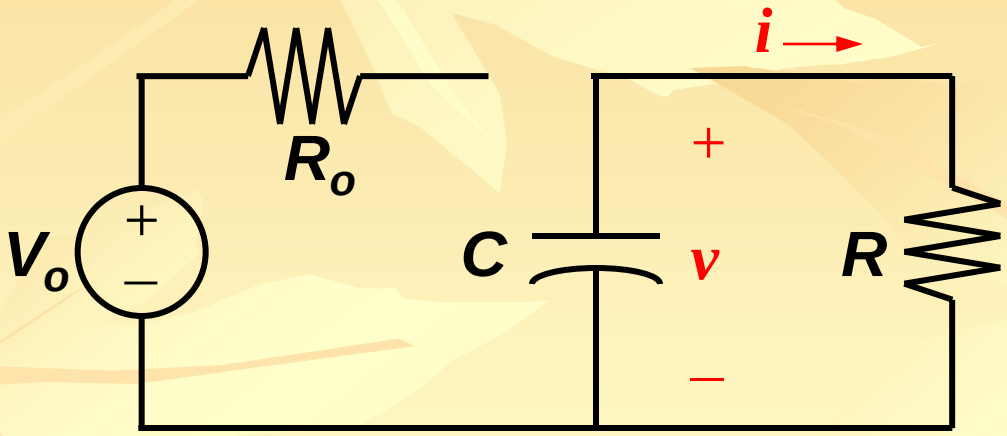


Notation:

- 0^- is used to denote the time just prior to switching
- 0^+ is used to denote the time immediately after switching.

Solving for the voltage *(t ≥ 0)*

- For $t \leq 0$, $v(t) = V_o$
- For t > 0, the circuit reduces to



Continue

- Applying KCL to the RC circuit:

$$i_C + i_R = 0 \longrightarrow (1)$$

$$C\frac{dv(t)}{dt} + \frac{v(t)}{R} = 0 \longrightarrow (2)$$

$$\frac{dv(t)}{dt} + \frac{v(t)}{RC} = 0 \longrightarrow (3)$$

$$\frac{dv(t)}{dt} = -\frac{v(t)}{RC} \longrightarrow (4)$$

$$\frac{dv(t)}{v(t)} = -\frac{1}{RC}dt \longrightarrow (5)$$

Continue

- From equation (5), let say:

$$\frac{dx}{x} = -\frac{1}{RC}dy \longrightarrow \textbf{(6)}$$

- Integrate both sides of equation (6):

$$\int_{V_o}^{v(t)} \frac{1}{x}du = -\frac{1}{RC}\int_0^t dy \longrightarrow \textbf{(7)}$$

- Therefore:

$$\ln\frac{v(t)}{V_o} = -\frac{t}{RC} \longrightarrow \textbf{(8)}$$

Continue

- Hence, the voltage is:

$$v(t) = v(0)e^{-t/RC} = V_o e^{-t/RC}$$

- Using Ohm's law, the current is:

$$i(t) = \frac{v(t)}{R} = \frac{V_o}{R} e^{-t/RC}$$

Continue

- The power dissipated in the resistor is:

$$p(t) = vi_R = \frac{V_o^2}{R} e^{-2t/RC}$$

- The energy absorb by the resistor is:

$$w = \frac{1}{2} CV_o^2 (1 - e^{-2t/RC})$$

Continue

- The time constant for the RC circuit equal the product of the resistance and capacitance,

- Time constant, $\tau = RC$ sec

- The expressions for voltage, current, power and energy using time constant concept:

$$v(t) = V_o e^{-t/\tau}$$

$$i(t) = \frac{V_o}{R} e^{-t/\tau}$$

$$p(t) = \frac{V_o^2}{R} e^{-2t/\tau}$$

$$w(t) = \frac{1}{2} C V_o^2 (1 - e^{-2t/\tau})$$

- For the case of capacitor, two important observation can be made,

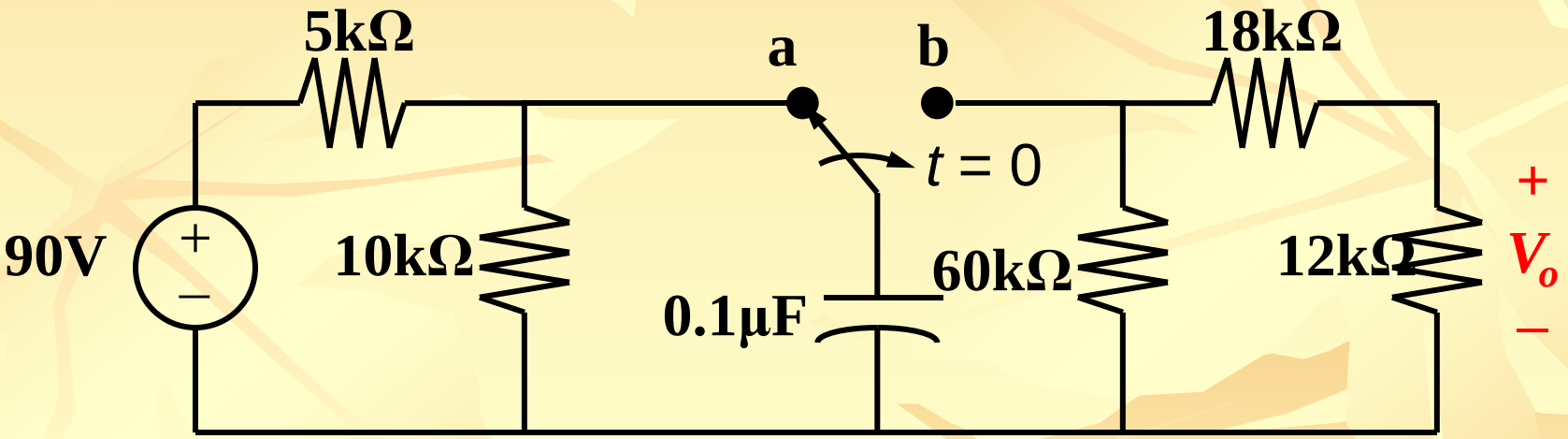
  1) capacitor behaves like an open circuit when being supplied by dc source

  (From, $i_c = C dv/dt$, when v is constant, $dv/dt = 0$. When current in circuit is zero, the circuit is open circuit.)

  2) in capacitor, the voltage is continuous / stays the same that is, $V_c(0^+) = V_c(0^-)$

Example

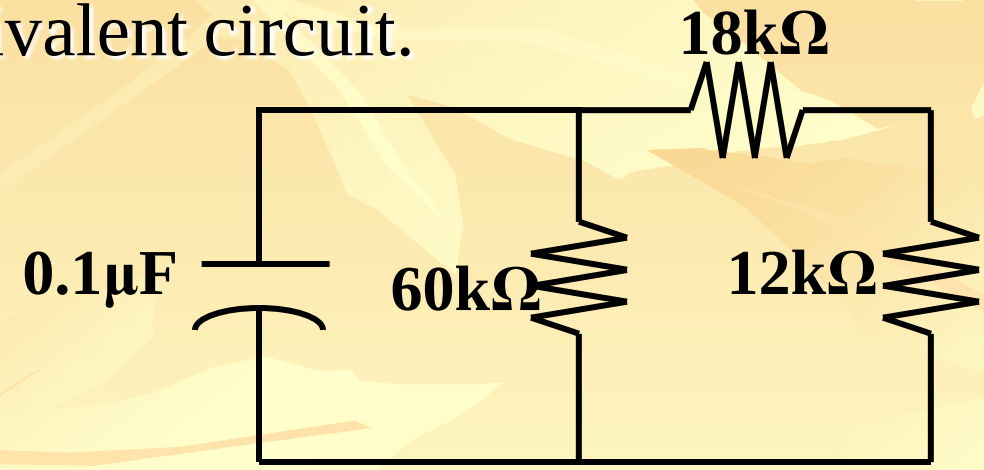
The switch has been in position *a* for a long time. At Time t = 0, the switch moves to *b*. Find the expressions for the $v_c(t)$, $i_c(t)$ and $v_o(t)$ and hence sketch them for t = 0 to t = 5τ.



Solution

Step 1:

Find t for t > 5τ that is when the switch was at *a*. Draw the equivalent circuit.

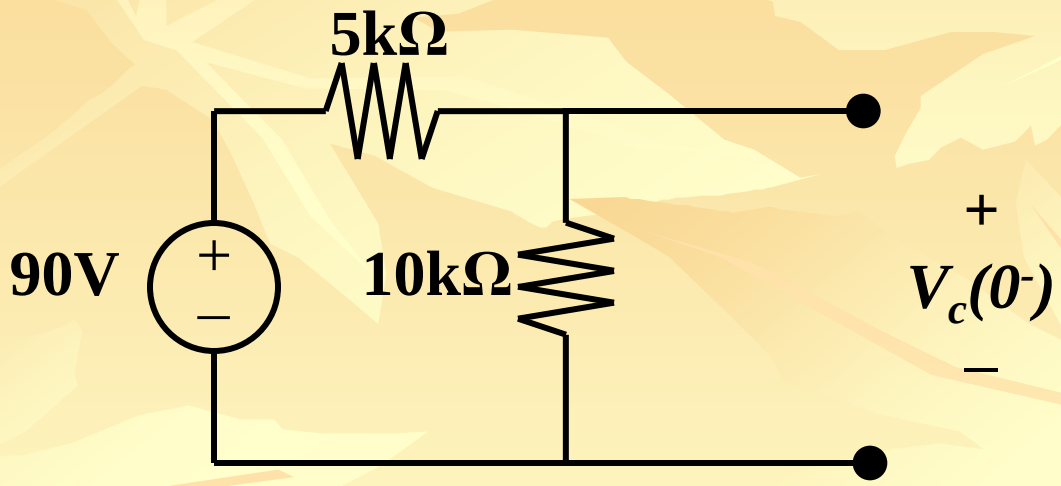


$$R_T = (18k\Omega + 12k\Omega) // 60k\Omega = 20k\Omega$$

$$\tau = R_T C = 20 \times 10^3 \times 0.1 \times 10^{-6} = 2ms$$

Step 2:

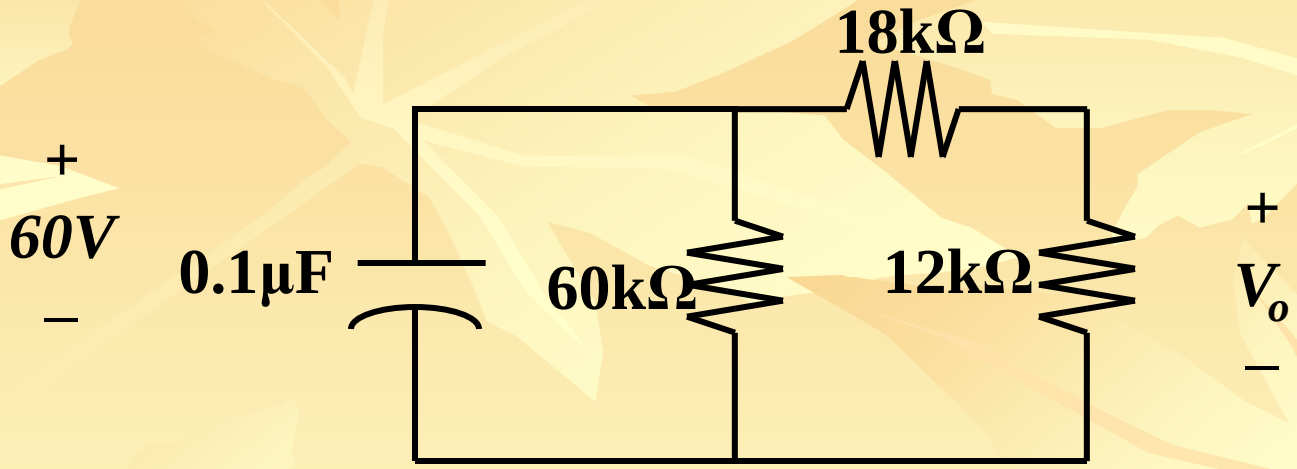
At t = 0, the switch was at *a*. the capacitor behaves like An open circuit as it is being supplied by a constant source.



$$v_c(0^-) = \frac{10}{15} \times 90 = 60V$$

Step 3:

At t = 0^+, the instant when the switch is at *b*.



The voltage across capacitor remains the same at this particular instant.

$$v_c(0^+) = v_c(0^-) = 60V$$

Using voltage divider rule,

$$V_o(0^+) = \frac{12}{30} \times 60 = 24V$$

Hence;

$$v_c(t) = 60e^{-500t}V$$

$$v_o(t) = 24e^{-500t}V$$

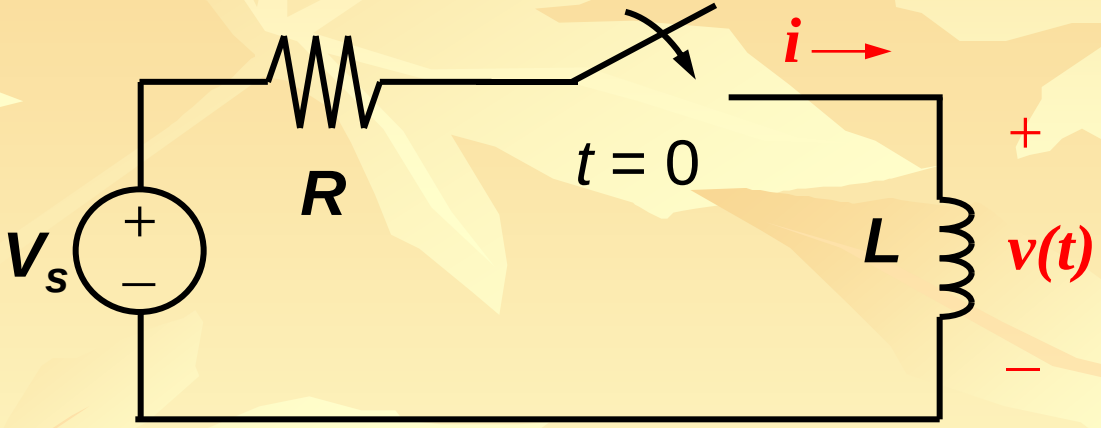
$$i_c(t) = -0.03e^{-500t}A$$

Summary

No	RL circuit	RC circuit
1	$\tau = \frac{L}{R}$	$\tau = RC$
2	Inductor behaves like a short circuit when being supplied by dc source for a long time	Capacitor behaves like an open circuit when being supplied by dc source for a long time
3	Inductor current is continuous $i_L(0^+) = i_L(0^-)$	Voltage across capacitor is continuous $v_C(0^+) = v_C(0^-)$

Step Response of RL Circuit

- The switch is closed at time t = 0.



- After switch is closed, using KVL

$$V_s = Ri(t) + L\frac{di}{dt} \longrightarrow \textbf{(1)}$$

Continue

- Rearrange the equation;

$$\frac{di(t)}{dt} = \frac{-Ri(t)+V_s}{L} = \frac{-R}{L}\left(i(t) - \frac{V_s}{R}\right) \longrightarrow \textbf{(2)}$$

$$di = \frac{-R}{L}\left(i - \frac{V_s}{R}\right)dt \longrightarrow \textbf{(3)}$$

$$\frac{-R}{L}dt = \frac{di}{i(t) - V_s/R} \longrightarrow \textbf{(4)}$$

$$-\frac{R}{L}\int_0^t dv = \int_0^{i(t)} \frac{du}{u - (V_s/R)} \longrightarrow \textbf{(5)}$$

Continue

- Therefore:

$$-\frac{R}{L}t = \ln\frac{i(t)-(V_s/R)}{I_0-(V_s/R)} \longrightarrow \textbf{(5)}$$

- Hence, the current is;

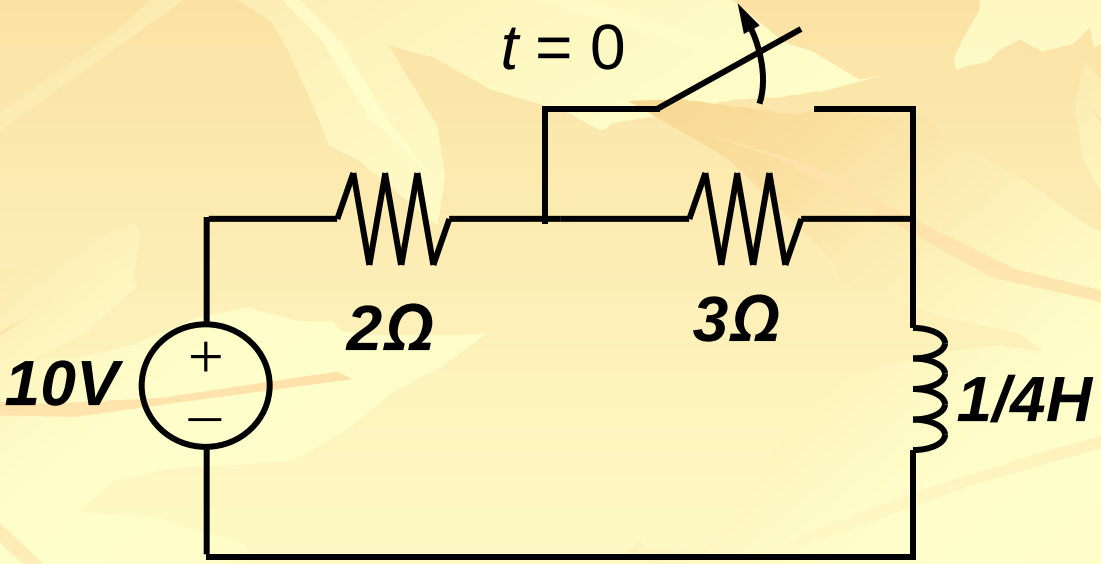
$$i(t) = \frac{V_s}{R} + \left(I_o - \frac{V_s}{R}\right)e^{-(R/L)t}$$

- The voltage;

$$v(t) = (V_s - I_o R)e^{-(R/L)t}$$

Example

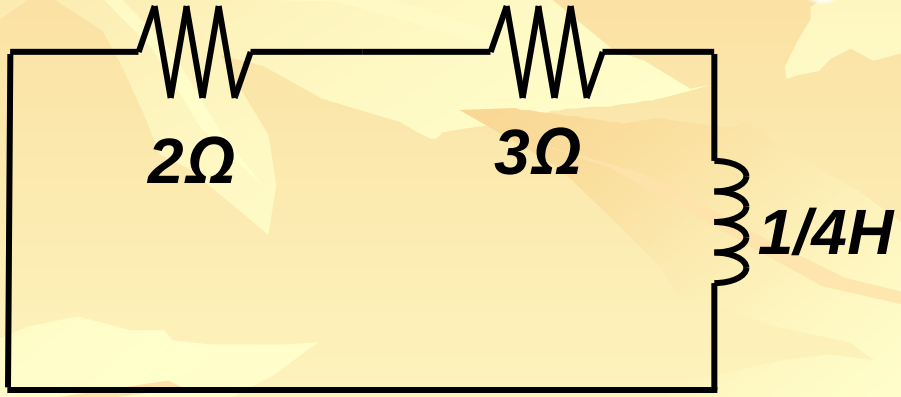
The switch is closed for a long time at t = 0, the switch opens. Find the expressions for $i_L(t)$ and $v_L(t)$.



Solution

Step 1:

Find τ for t > 0. The switch was opened. Draw the equivalent circuit. Short circuit the voltage source.



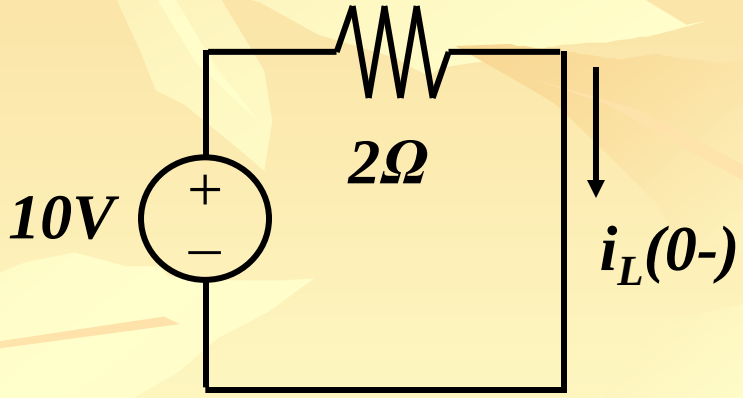
$$R_T = (2+3)\Omega = 5\Omega$$

$$\tau = \frac{L}{R_T} = \frac{1}{20}s$$

Continue

Step 2:

At t = 0^-, the switch was closed. Draw the equivalent circuit with 3Ω shorted and the inductor behaves like a short circuit.



$$i_L(0^-) = 10/2 = 5A$$

Continue

Step 3:

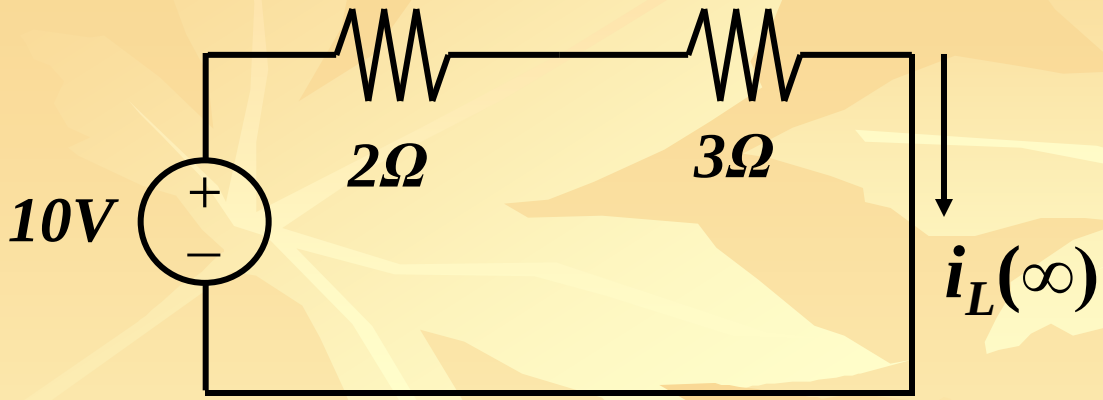
At t = 0^+, the instant switch was opened. The current in inductor is continuous.

$$I_0 = i_L(0^+) = i_L(0^-) = 5A$$

Step 4:

At t =∞, that is after a long time the switch has been left opened. The inductor will once again be behaving like a short circuit.

Continue



$$i_L(\infty) = V_s / R_T = 2A$$

Hence:

$$i_L(t) = \frac{V_s}{R} + \left(I_o - \frac{V_s}{R} \right) e^{-(R/L)t}$$

$$\boxed{i_L(t) = 2 + 3e^{-20t} A}$$

Continue

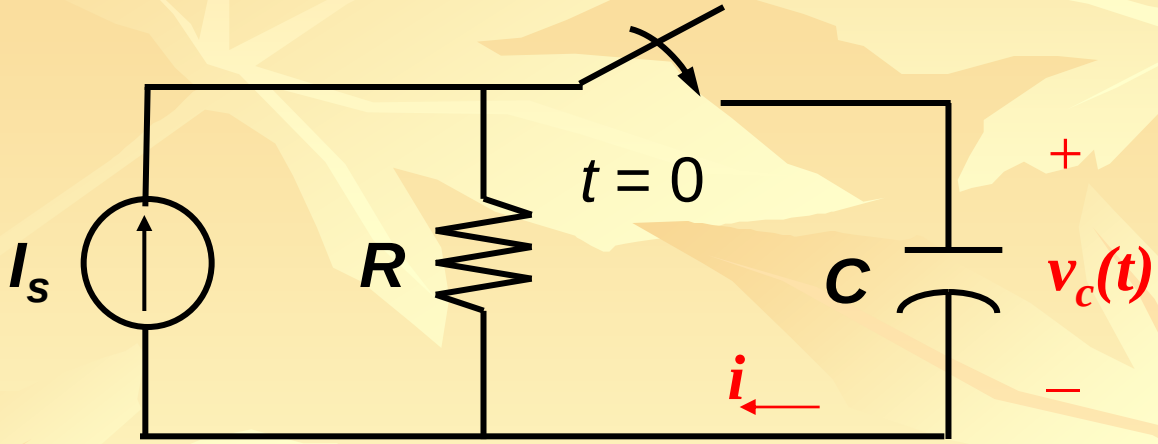
- And the voltage is:

$$v_L(t) = (V_s - I_o R)e^{-(R/L)t}$$

$$\boxed{v_L(t) = -15e^{-20t}V}$$

Step Response of RL Circuit

- The switch is closed at time t = 0



- From the circuit;

$$I_s = C\frac{dv_c}{dt} + \frac{v_c}{R} \longrightarrow \textbf{(1)}$$

Continue

- Division of Equation (1) by C gives;

$$\frac{I_s}{C} = \frac{dv_c}{dt} + \frac{v_c}{RC} \longrightarrow \textbf{(2)}$$

- Same mathematical techniques with RL, the voltage is:

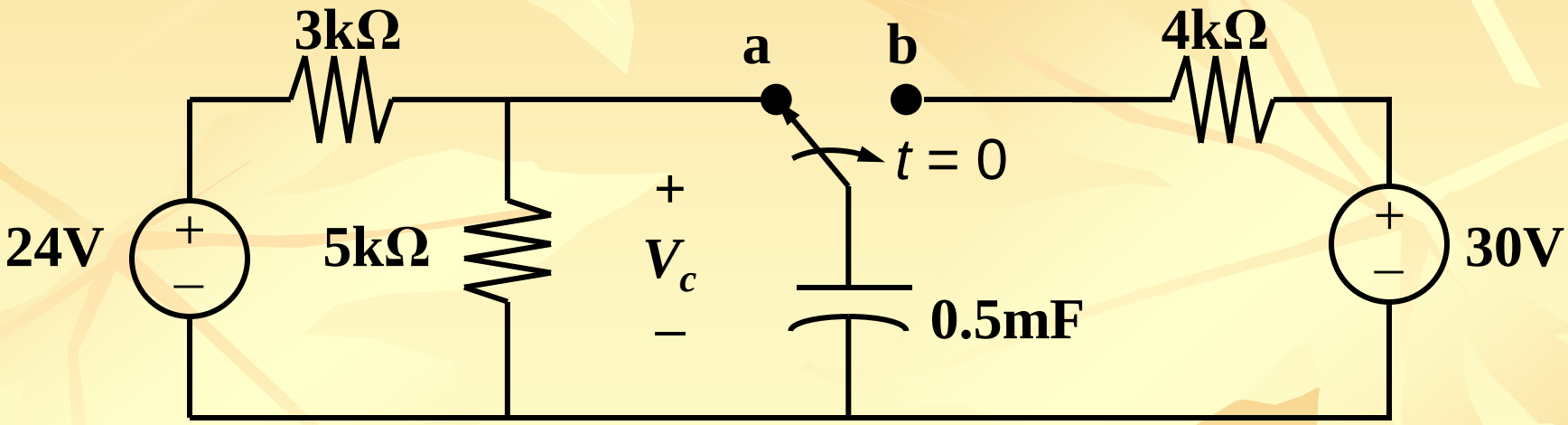
$$v_c(t) = I_s R + (V_o - I_s R)e^{-t/RC}$$

- And the current is:

$$i(t) = \left(I_s - \frac{V_o}{R} \right) e^{-t/RC}$$

Example

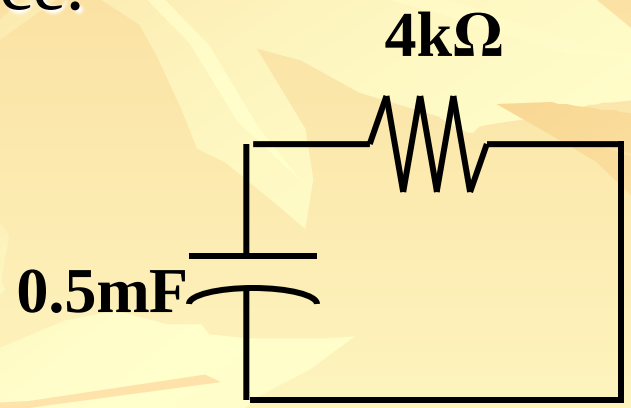
The switch has been in position *a* for a long time. At t = 0, the switch moves to *b*. Find $V_c(t)$ for t > 0 and calculate its value at t = 1s and t = 4s.



Solution

Step 1:

To find τ for t > 0, the switch is at *b* and short circuit the voltage source.

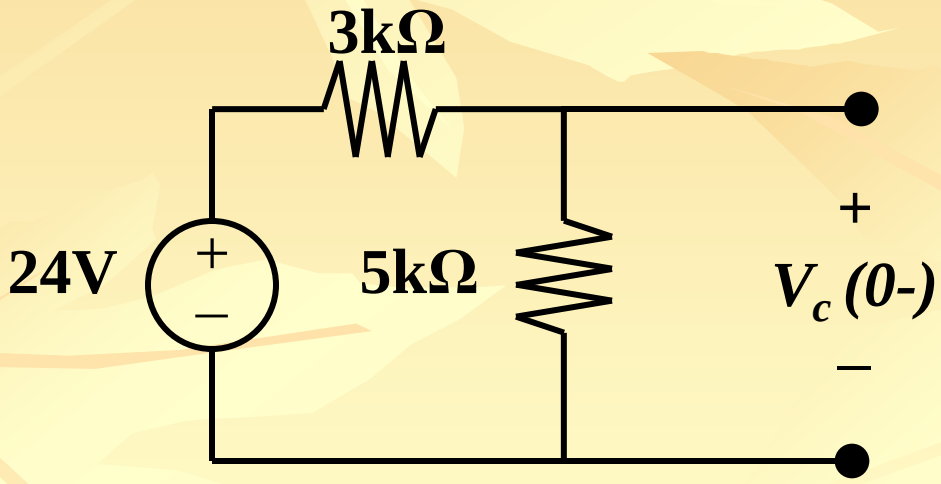


$$\tau = RC = 2s$$

Continue

Step 2:

The capacitor behaves like an open circuit as it is being supplied by a constant dc source.



From the circuit,

$$V_c(0^-) = 24 \times \frac{5}{8} = 15V$$

Continue

Step 3:

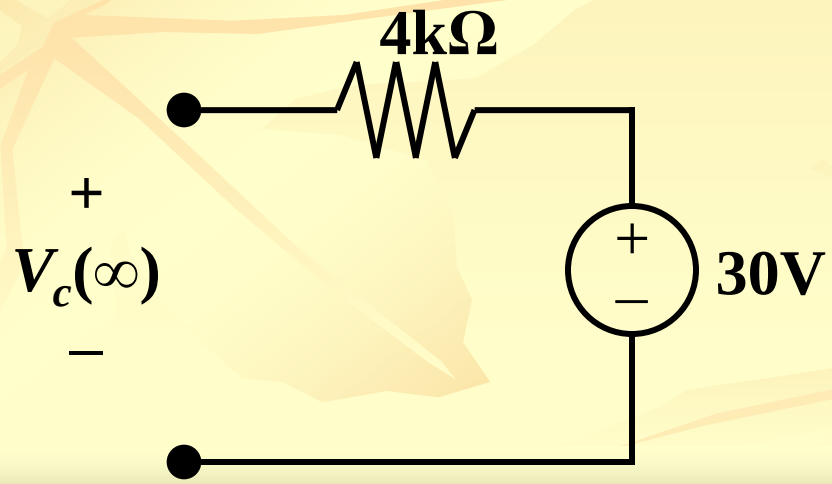
At t = 0^+, the instant when the switch is just moves to *b*.

Voltage across capacitor remains the same.

$$V_c(0^+) = V_c(0^-) = 15V$$

Step 4:

At t = ∞, the capacitor again behaves like an open circuit since it is being supplied by a constant source.



$$V_c(\infty) = 30V$$

Continue

Step 5:

Hence,

$$V_c(t) = 30 + (15 - 30)e^{-0.5t} = 30 - 15e^{-0.5t} V$$

At t = 1s, $V_c(t)$ = 20.9V

At t = 4s, $V_c(t)$ = 28 V

THE END